

Supplement 2: Selecting negative control genes for RUV in placental transcriptome studies across gestational ages

Motivation

Supplement 1 identified RUV (Remove Unwanted Variation) as one of the most promising alternatives to ComBat for integrating microarray studies whose biological groups are not balanced across batches [1, 2]. Unlike ComBat, which estimates batch parameters from all genes, RUV estimates the unwanted factors of variation from a set of *negative control genes* — genes assumed *a priori* not to be affected by the biological condition of interest. This makes RUV robust to confounding between batch and biology, because the unwanted factors are learned exclusively from genes that should carry no biological signal.

The quality of the correction depends almost entirely on the quality of the control gene set. If genuine differentially expressed genes leak into the controls, RUV will treat part of the biological signal as unwanted variation and remove it, deflating true effect sizes. Conversely, if the controls contain genes that are affected by technical artefacts in a non-representative way, RUV will under-correct.

For the placental transcriptome, selecting controls is unusually difficult. Mikheev et al. showed that roughly 25% of protein-coding genes are differentially expressed between first trimester and term placenta [6], meaning that a naive random sample of “housekeeping” genes has a high probability of containing genuinely regulated transcripts.

This supplement reviews three strategies for assembling a negative control gene list and recommends a combined approach for the pipeline.

1 What are negative control genes?

In the RUV framework, the observed expression matrix Y is modelled as

$$Y = X\beta + W\alpha + \varepsilon,$$

where $X\beta$ captures the biology of interest (e.g. trimester contrasts), $W\alpha$ captures k unwanted factors (batch, platform, RNA quality), and ε is noise. The matrix W is unobserved and must be estimated.

RUVg [2] estimates W by performing a singular value decomposition on the subset of rows of Y that correspond to the negative control genes. Because these genes are assumed to have $\beta = 0$ for the contrast of interest, any systematic variation among them is attributed to W . The estimated $\hat{W}$ is then included as a covariate in the downstream differential expression model, absorbing batch effects without removing biology.

The control genes need not be truly invariant in an absolute sense; they must merely be *unaffected by the specific contrast being tested*. A gene that varies with BMI but not with gestational age is a valid control for a trimester comparison.

2 Strategy 1: Curated housekeeping gene lists

2.1 The Eisenberg–Levanon list

Eisenberg and Levanon [3] compiled a list of 3,804 human genes that are expressed uniformly across 16 normal tissues in RNA-seq data (Human BodyMap 2.0). Selection criteria included low coefficient of variation across tissues and ubiquitous detection (RPKM > 1 in all tissues). The list is publicly available as Entrez Gene IDs and RefSeq accessions.

Strengths. The list is large, well-documented, and immediately usable: it can be intersected with the gene universe of the merged expression matrix without any additional experiment. Its size (typically 2,000–3,000 genes survive after mapping to the microarray platform) provides statistical stability for estimating W .

Limitations. The list was derived from adult tissues and does not include placenta. Genes that are housekeeping in liver and brain may be dynamically regulated during placental development. Furthermore, “uniform across tissues” does not guarantee “uniform across gestational ages within one tissue.”

2.2 Other curated lists

Earlier compilations by Warrington et al. [7] and de Jonge et al. [8] are smaller (500–600 genes) and based on older microarray platforms. They are largely subsets of the Eisenberg–Levanon list and add little independent information for this application.

3 Strategy 2: Empirical (in-silico) control genes

Instead of relying on an external list, one can select control genes empirically from the data itself. The standard recipe is:

1. Run a preliminary differential expression analysis (e.g. limma with no batch correction, or limma with batch as a fixed covariate) on the merged matrix.
2. Rank genes by their raw or adjusted p -value in *descending* order (least significant first).
3. Take the top n genes (e.g. the 1,000 genes with the highest p -values) as empirical negative controls.

This approach was proposed by Gagnon-Bartsch and Speed [1] and refined by Risso et al. [2] in the RUVSeq package (the “RUVr” and “RUVg with empirical controls” variants).

Strengths. The controls are data-driven and automatically adapted to the specific contrast. No external annotation is needed.

Limitations. The procedure is circular: the preliminary analysis is confounded by the very batch effects that RUV is meant to remove. If batch effects push a truly non-DE gene to low p -values (false positive), it will be excluded from the controls; if they push a truly DE gene to high p -values (false negative), it will be included. In practice, mild circularity is tolerable when the batch effect is not too large relative to the biological signal, but it is a concern for the placental collection where batch and trimester are partially aliased.

Jacob et al. [9] showed that RUV with empirical controls can still inflate false discoveries when the confounding is severe, and recommended combining empirical selection with an external list.

4 Strategy 3: Placenta-validated reference genes

qRT-PCR normalisation studies have specifically evaluated reference gene stability across gestational ages in human placenta.

4.1 Meller et al. (2005)

Meller et al. [4] used geNorm and NormFinder to rank ten candidate reference genes across first, second, and third trimester placentas. Their key findings:

- **TBP** (TATA-binding protein), **SDHA** (succinate dehydrogenase A), and **YWHAZ** (tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein zeta) were the three most stable genes.
- The commonly used references **ACTB** (β -actin) and **GAPDH** were among the *least* stable, varying significantly between trimesters.
- The authors recommended using at least two of the top three genes as qPCR normalisers for gestational-age comparisons.

4.2 Drewlo et al. (2012)

Drewlo et al. [5] re-examined reference gene stability in first-trimester versus term placenta using a broader panel. They confirmed TBP and YWHAZ as stable and additionally identified **TOP1** (topoisomerase I) and **HPRT1** as reliable across early and late gestation. They also confirmed that GAPDH is unsuitable for trimester comparisons.

Strengths. These genes have been experimentally validated in the exact tissue and developmental window of interest. Even a handful of such genes can anchor the RUV estimate.

Limitations. The validated sets are very small (3–6 genes), far fewer than the hundreds to thousands recommended for stable RUV factor estimation [2]. They cannot be used as the sole control set but serve as high-confidence anchors.

5 Empirical validation against our pipeline results

We evaluated the literature-recommended reference genes against differential expression results from ten Phase 2B pipeline runs spanning two contrasts (First vs. Second Trimester and Second Trimester vs. Term), multiple dataset selections, and two batch correction strategies (ComBat and batch-in-limma). Table 1 summarises the log fold change and adjusted p -value for each gene, reported as the range across runs for each contrast.

Several observations emerge:

- **YWHAZ** is the only recommended gene that is truly stable across all runs and both contrasts (FDR > 0.12 everywhere, $\log FC \approx 0$). It is the strongest single anchor gene for RUV in this dataset.
- **HPRT1** was validated by Drewlo et al. specifically for the *first trimester vs. term* comparison, and our 1.2 results are consistent with that finding (FDR > 0.10). However, HPRT1 is *strongly* differentially expressed in the Second Trimester vs. Term contrast ($\log FC = -0.88$, FDR = 2.5×10^{-13} – 5.3×10^{-6}), a pairwise comparison that Drewlo et al. did not test separately. This finding is consistent across all five 2.3 runs, including the conservative batch-in-limma approach, ruling out a batch-effect artefact. HPRT1 must be excluded from the negative control set for the 2.3 contrast.

Table 1: All candidate reference genes from Meller et al. [4] (M, tested across all three trimesters) and Drewlo et al. [5] (D, tested first vs. third trimester) evaluated against Phase 2B pipeline results. logFC and FDR are median logFC and minimum adj. p -value across all runs for each contrast.

Gene	Source	Literature verdict	1st vs. 2nd		2nd vs. Term	
			logFC	FDR	logFC	FDR
<i>Top stable (recommended as references)</i>						
YWHAZ	M, D	top 3 stable (M); stable (D)	+0.03	0.77	+0.12	0.12
TBP	M, D	top 3 stable (M); stable (D)	+0.05	0.11	+0.20	0.002
SDHA	M	top 3 stable (M)	<i>Not in gene universe</i>			
TOP1	D	stable (D)	-0.30	4×10^{-4}	-0.02	0.29
HPRT1	D	stable (D)	+0.16	0.11	-0.88	2×10^{-15}
RPL13A	D	stable (D)	-0.07	0.13	-0.02	0.51
IPO8	D	stable (D)	+0.11	0.12	-0.06	0.58
<i>Middle rank</i>						
UBC	M	mid-rank (M)	<i>Not in gene universe</i>			
B2M	M	mid-rank (M)	+0.24	0.003	-0.13	0.03
HMBS	M	mid-rank (M)	-0.39	0.005	-0.66	3×10^{-11}
PPIA	M	mid-rank (M)	+0.01	0.43	-0.63	3×10^{-10}
POLR2A	D	tested (D)	-0.11	0.001	-0.03	0.45
PGK1	D	tested (D)	-0.28	4×10^{-4}	-0.09	0.02
GUSB	D	tested (D)	+0.14	0.10	+0.50	2×10^{-7}
CYC1	D	tested (D)	-0.09	0.19	-0.22	0.001
EIF4A2	D	tested (D)	+0.03	0.048	+0.19	0.004
<i>Flagged as unstable (avoid)</i>						
GAPDH	M, D	least stable (M); unstable (D)	-0.29	4×10^{-4}	-0.58	1×10^{-6}
ACTB	M	least stable (M)	<i>Not in gene universe</i>			
RPLP0	D	tested (D)	<i>Not in gene universe</i>			

- **TBP** is stable in the 1_2 contrast but reaches significance in some 2_3 runs (FDR = 0.003), albeit with a small effect size (logFC = +0.20). It is borderline and contrast-dependent.
- **TOP1** shows the reverse pattern: mildly significant in some 1_2 runs (FDR = 0.0004) but completely stable in 2_3.
- **GAPDH** is confirmed unstable in both contrasts, as predicted by the literature.
- The qPCR validation studies used small sample sizes and did not test each pairwise trimester contrast separately. Our results across ten pipeline configurations reveal *contrast-specific instability* that the original studies could not detect.

These findings reinforce the recommended strategy of intersecting curated lists with empirical invariance filtering (Section 5), and of performing this filtering separately for each contrast.

6 Recommended strategy and final gene lists

We constructed contrast-specific negative control gene lists by applying a five-rule selection procedure to the Eisenberg–Levanon housekeeping gene catalogue. The procedure was applied independently for each trimester contrast (First vs. Second Trimester and Second Trimester vs. Term), because a gene that is stable in one comparison may be differentially expressed in the other (cf. HPRT1 above).

6.1 Starting set

The Eisenberg–Levanon list of 3,804 human housekeeping gene symbols was mapped to NCBI Entrez Gene IDs via `org.Hs.eg.db`, yielding 3,505 unique protein-coding genes.

6.2 Selection rules (applied per contrast)

For each gene and each contrast, five rules are evaluated in order. The first matching rule determines the decision.

1. **Exclude if DEG in primary runs.** A gene is excluded if it is differentially expressed (adj. $P < 0.05$) in the *balanced* run (only datasets contributing samples to both groups; ComBat batch correction) **or** the *batch-in-limma* run (all 14 datasets; batch included as a covariate in the limma design matrix, no ComBat). Both are conservative approaches: the balanced run uses only datasets with samples on both sides of the contrast (fewer datasets, less statistical power), and batch-in-limma avoids ComBat entirely (producing fewer DEGs overall). If even these conservative analyses flag a gene as DE, it is likely genuinely regulated rather than a batch artefact.
2. **Exclude if absent from primary runs but always DEG elsewhere.** If a gene is not present in either primary run (filtered by coverage or variance) but *is* a DEG in every other run where it passes filters (all-datasets ComBat, restoration, and second-trimester-only runs), it is excluded. Rationale: there is no evidence of stability, only evidence of differential expression.
3. **Literature stable override.** If placenta-specific qPCR studies (Meller [4], Drewlo [5], Lanoix [10], St-Pierre [11], Murthi [12]) rank the gene as *stable*, it is included even if it is a weak DEG in our pipeline — **unless** our data shows it is a strong DEG ($|\log\text{FC}| > 0.5$ and adj. $P < 0.05$) in a primary run. This protects literature-validated genes from marginal false positives while still excluding those with clear biological signal.

4. **Literature unstable override.** If the literature flags the gene as *unstable* in placenta (e.g. ACTB, GAPDH, RPLP0), it is excluded regardless of our empirical results. These genes have been independently shown to vary across gestational ages or pathological conditions in placental tissue.
5. **Default to empirical data.** Genes with no literature verdict (the vast majority of the 3,505 Eisenberg–Levanon genes) follow the empirical evidence: included if non-DEG in all runs where present, excluded if DEG or absent from all runs.

6.3 Additional literature sources

Beyond Meller and Drewlo, we consulted three additional placental qPCR studies to inform Rules 3 and 4:

- **Lanoix et al. (2012)** [10] evaluated nine reference genes (ACTB, GAPDH, HPRT1, PPIA, SDHA, TBP, TOP1, YWHAZ, 18S rRNA) in normotensive vs. preeclamptic and normotensive vs. gestational diabetes placentas at term, using both geNorm and NormFinder. HPRT1 and PPIA were the most stable in preeclampsia; YWHAZ and PPIA in GDM. Rankings differed substantially between the two disease contexts, underscoring the importance of contrast-specific evaluation.
- **St-Pierre et al. (2017)** [11] tested 28 reference gene candidates in 20 term placentas (10 male, 10 female) and reported that TBP and YWHAZ form the best geNorm pair, while IPO8 ranks highest by NormFinder. They also identified sex-biased expression for HPRT1, GUSB, RPL13A, and RPS18 (significantly lower Cq values in male placentas), relevant because our pipeline includes fetal sex as a covariate.
- **Murthi et al. (2008)** [12] evaluated six genes in normal vs. fetal growth restriction (FGR) placentas. GAPDH, 18S rRNA, and YWHAZ were the most stable; ACTB and TBP the least stable — notably the opposite ranking from Meller’s gestational-age study, further illustrating context-dependence.

6.4 Results

Table 2: Final negative control gene list sizes after applying the five-rule selection procedure to 3,505 Eisenberg–Levanon genes.

Contrast	Included	Excluded	Total EL
First vs. Second Trimester (1_2)	2,728	777	3,505
Second Trimester vs. Term (2_3)	1,948	1,557	3,505
Both contrasts (intersection)	1,597	—	—

The 2.3 contrast yields a smaller control set because substantially more genes are differentially expressed between the second trimester and term (a larger developmental change than first to second trimester). Both lists are well above the several-hundred-gene minimum recommended for stable estimation of the unwanted factors W [2, 1]. The intersection of 1,597 genes stable in *both* contrasts is available for analyses that require a single unified control set.

7 Choosing k and diagnostics

The number of unwanted factors k controls the aggressiveness of the correction. Too few factors leave residual batch effects; too many absorb biological signal.

7.1 Selecting k

1. Compute the singular values of the control-gene submatrix.
2. Plot the scree (proportion of variance explained per factor).
3. Choose k at the elbow, where the curve flattens.
4. Cap k at $b - 1$, where b is the number of batches (datasets), to avoid over-fitting.

For the placental collection with 4–8 datasets, k typically falls in the range 2–5.

7.2 Diagnostic checks

After applying RUV with the chosen k :

- **PCA.** The first two principal components of the corrected matrix should separate by trimester, not by dataset. Compare to the uncorrected and ComBat-corrected PCAs.
- **RLE (Relative Log Expression) plots.** Per-sample median deviations should be centred near zero and have similar spread across datasets.
- **DEG overlap.** Compare the DEG list from RUV-corrected limma to the ComBat-corrected list. Genes significant under both methods are high-confidence; genes appearing only under one method warrant closer inspection.
- **Control gene expression after correction.** The negative control genes should show no systematic dataset-to-dataset shifts in the corrected matrix. A persistent shift indicates under-correction (k too small).

8 Empirical evaluation of RUV batch correction

We applied the contrast-specific negative control gene lists from Section 5 to the Phase 2B pipeline and compared the RUV-corrected results against ComBat. Three RUV variants were tested:

1. **RUVg** with fixed k ($k = 2, 3, 4$): SVD on the centred control-gene submatrix yields $\hat{W}$, which is included as covariates in the limma design (the standard RUVg + limma workflow). The expression matrix is not altered; batch correction occurs implicitly through the linear model.
2. **RUVinv** [13]: An inverse-regression variant from the **ruv** R package that integrates over all possible k rather than requiring a fixed choice. RUVinv performs its own differential expression test (bypassing limma) and returns $\hat{\beta}$ (log fold change), t -statistics, and p -values directly.

Each was run on both the balanced configuration (only datasets contributing samples to both groups) and the all-datasets configuration (all 14 datasets), for both contrasts. Only the “none” (no imputation) results are discussed here, because the imputed matrices amplify the same patterns.

8.1 DEG counts

Table 3 compares the number of differentially expressed genes (adj. $P < 0.05$ and $|\log\text{FC}| > 1$) across methods.

Table 3: Significant DEGs (FDR < 0.05 and $|\log\text{FC}| > 1$) for ComBat and RUV variants, without imputation.

	ComBat	RUVg $k=2$	RUVg $k=3$	RUVg $k=4$	RUVinv
1.2 balanced	227	267	163	81	10
2.3 balanced	529	534	428	443	364
1.2 all datasets	189	701	643	238	10
2.3 all datasets	435	1,561	1,364	1,011	284

8.2 Observations

Balanced runs: RUVg $k=2$ matches ComBat. When only balanced datasets are used (2–4 datasets per contrast), RUVg with $k = 2$ produces DEG counts close to ComBat: 267 vs. 227 for the 1.2 contrast and 534 vs. 529 for 2.3. The overlap between the two gene lists is substantial ($> 85\%$ of the smaller set), suggesting that both methods capture the same biological signal when batch–biology confounding is minimal.

All-datasets runs: RUVg inflates DEGs. With all 14 datasets, RUVg $k = 2$ detects far more DEGs than ComBat: 701 vs. 189 (1.2, a $3.7\times$ inflation) and 1,561 vs. 435 (2.3, $3.6\times$). PCA of the RUVg-corrected matrix reveals persistent dataset clustering, indicating that $k = 2$ is insufficient to capture the batch structure across 14 studies. The residual batch variation is absorbed by the biology term in the limma model, producing spurious differential expression.

Higher k is not a simple fix. Increasing k to 3 or 4 reduces the inflation but introduces instability. For the 1.2 all-datasets contrast, $k = 3$ still yields 643 DEGs ($3.4\times$ ComBat) while $k = 4$ drops to 238 (closer to ComBat’s 189). For the balanced runs, higher k reduces DEGs below ComBat levels (81 at $k = 4$ vs. 227 ComBat for 1.2 balanced), indicating over-correction — the unwanted factors absorb biological signal. PCA confirms this: trimester separation deteriorates with increasing k in balanced runs. There is no single k that works well for both balanced and all-datasets configurations.

RUVinv is conservative to the point of under-detection. RUVinv produces only 10 DEGs for both 1.2 configurations (balanced and all-datasets alike), despite ComBat detecting 227 and 189 respectively. For the 2.3 contrast (where the biological signal is stronger), RUVinv recovers 364 (balanced) and 284 (all-datasets) DEGs, closer to ComBat’s 529 and 435 but still below. The integration over all k appears to heavily penalise effect estimates in the presence of moderate batch–biology confounding, collapsing fold changes toward zero.

8.3 Why RUV underperforms in this setting

The core difficulty is *partial confounding between batch and biology*. In the all-datasets configuration, several GEO series contribute samples from only one trimester group, creating a structural correlation between dataset identity and gestational age. This violates the key assumption of RUVg: that the unwanted factors W are orthogonal to the biology X . When they are not, the estimated $\hat{W}$ captures a mixture of batch effects and biological signal, and including it in the limma design absorbs part of both.

ComBat explicitly models batch as a categorical factor with known labels and uses an empirical Bayes shrinkage prior, which stabilises the estimates even when batches are partially confounded with the groups of interest. RUVg, by contrast, estimates W blindly from the control-gene submatrix without knowing the batch structure, making it more vulnerable to confounding.

Additionally, k selection is fundamentally ill-posed in this setting. The rule of thumb $k \leq b - 1$ (where b is the number of batches) suggests k could be as high as 13 for 14 datasets, but the degrees of freedom consumed by such a large W would eliminate most statistical power. The scree plot of the control-gene submatrix shows no clear elbow, because the variance contributions of batch and biology are entangled in the same singular vectors.

8.4 Possible improvements

Several strategies could improve RUV performance in this setting:

1. **Two-stage correction.** Apply ComBat first to remove gross batch effects, then use RUV on the residuals to capture subtler unwanted variation (e.g., platform differences, RNA quality) that ComBat does not model. This avoids asking RUV to handle a confounding structure it is not designed for.
2. **Estimate W from balanced subsets only.** Restrict the SVD that produces $\hat{W}$ to samples from balanced datasets (where both groups are represented), then project $\hat{W}$ onto the full sample set. This decouples the factor estimation from the confounded structure.
3. **RUV4 [13].** The RUV4 variant uses residuals from a first-pass regression that includes the biology term X , estimating W from what remains after removing the signal of interest. This is theoretically more robust to batch–biology confounding than RUVg, though it still requires a reasonable first-pass model.
4. **RUV-III [14].** Uses technical replicate samples (or pseudo-replicates constructed from the same tissue type across studies) rather than control genes to estimate W . This sidesteps the question of whether the control gene list is appropriate, though our dataset does not contain formal technical replicates.
5. **Data-driven k selection.** Instead of the scree plot, select k by optimising a composite objective: maximise silhouette score for trimester separation while minimising dataset clustering in the corrected PCA space. This directly targets the dual goal of batch removal and biology preservation.
6. **Iterative control gene refinement.** Run RUV with an initial k , perform DE, update the control gene list by removing newly significant genes, and repeat until convergence. This addresses the possibility that some genes in the Eisenberg–Levanon list are genuinely DE in our specific context despite passing the pipeline-based filtering in Section 5.

Given that ComBat already performs well in both balanced and all-datasets configurations and that RUV’s advantage — robustness to batch–biology confounding — is offset by the difficulty of choosing k in this partially confounded design, we retain ComBat as the primary batch correction method for the main analysis. The RUV results are preserved as a secondary validation: genes detected by both ComBat and RUVg ($k = 2$, balanced configuration) represent a high-confidence subset.

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